Artificial Intelligence in Developing Assistive Devices for The Elderly The Smart Cane Model

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***Abstract*—With the rapid growth of artificial intelligence (AI), a new generation of innovative assistive devices has been developed for elderly people to enhance their mobility and safety, such the Smart Cane, a system built with a Long Short-Term Memory (LSTM) model to improve mobility, prevent falls, and provide real-time assistance. This research combines mobility tracking metrics, device feedback from the users, and system performance to demonstrate the impact of the Smart Cane. The outcome shows a significant increase in mobility, such (MII jumped from 50-65 to 75-90), 89% falls prevention accuracy, and a very high satisfaction score (4.6/5). The results indicate that LSTM assistive models may have profound implications for the independence and well-being of elderly individuals. Continuing technological development and cost-effectiveness remain prerequisites for maintenance and are potential future delivery methods made possible by IOT and continuous health monitoring, the study shows. This study adds to the emerging field of AI-driven mobility solutions and serves as a steppingstone towards continued progress in assistive technology.**

Keywords— LSTM, Assistive Technology, Smart Cane, Elderly Mobility, Fall Prevention, AI-Powered Devices, User Satisfaction, Independent Living, Healthcare Innovation, AI in Assistive Devices.

I. INTRODUCTION

As the world's elderly population grows, ensuring seniors can move around on their own as much as possible is an urgent challenge. Human strength degradation, excessive fatigue, and a challenge to maintain daily life are triggered by the age-related diseases such as arthritis, muscle weakness, or some neurological diseases, increasing the fall risk in elderly people. [1] Traditional mobility aids like canes and walkers provide some support but aren't smart enough to meet an individual's specific needs — much less alert caregivers in an emergency.

This creates a needs for assistive devices that can give physical companionship and help-to-move, but also offers safety and usability as well, combined with real-time monitoring. With this challenge in mind, artificial intelligence (AI) could provide an innovative and transformative answer [2]. Built-in AI features can transform assistive appliances, like mobility aids into intelligent companions that analyzing the user needs and identifies obstacles [3], provide navigation help and even send alerts in an emergency. We also devise a model in this study, named as Smart Cane Model, which uses cutting-edge technology in it as artificial intelligence-based sensors and machine learning algorithms for providing mobility assistance and also safety to the users.

This innovative device can identify environmental risks, monitor the user's vital signs, and deliver immediate input to avoid potential accidents. It is in-line with worldwide initiatives aimed at developing assistive technology to enhance independence and lessen the burden on caregivers and healthcare systems [4]. in addition to its benefits of AI-powered assistance such, regular mobility aids, focusing on the device's role to enhance safety and the overall user experience [5].

Next section is going to review of existing literature on AI in assistive technologies is presented, followed by an outline of the dataset and preprocessing techniques used in developing the model, where the methodology describes the experimentation setting, performance measures, and implementation. Finally, results are analyzed and discussed leading to a conclusion about the performance and prospects of AI-based assistive devices for the elderly.

II. LITERATURE REVIEW

Artificial Intelligence (AI)-powered assistive devices have completely transformed the elderly care sector by supplying them with advanced and more functional mobility solutions than canes and walkers. Researchers are targeting an aging population with new technologies, including development of numerous AI-driven tools, smarter mobility devices that can enhance safety, functionality, and independence.

Veras et al. [6] developed a methodical protocol for AI-based mobility techniques, identifies existing research lacunae, and accentuates the need for flexible technical solutions that can accommodate a wide range of mobility challenges, Their paper explains how AI-based solutions can serve as a bridge between assistive technology and personalized care of older adults.

Naayini et al. [7] focus on AI-powered assistive technologies designed specifically for individuals with visual impairments, demonstrating how machine learning algorithms can enhance navigation and mobility. Their findings indicate that AI-powered vision systems integrated into assistive devices can significantly improve the mobility experience of elderly users.

Similarly, Herath et al. [8] discuss AI approaches for functional mobility assessments in elderly healthcare, underscoring the importance of vision-attentive technology in identifying movement patterns, fall risks, and posture stability. These studies reinforce the necessity of AI in assistive devices, particularly for real-time mobility assessment and hazard detection. A growing body of research also emphasizes the Internet of Medical Things (IoMT) and its role in elderly care.

Vasudevan et al. [9] provided an overview of AI-powered IoMT applications for elderly adults in independent living environments, outlining how smart assistive devices may be integrated into connected health ecosystems. By equipping smart mobility aids with AI and IoMT capabilities, their work shows a way to improve

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health monitoring and provide emergency alerts in the case of the user falling or exhibiting unusual movement patterns.

These features were integrated into the assistive technologies, increasing older people's safety and independence. Other recent studies have looked at how smart canes evolved into multipurpose mobility devices. Messaoudi [10] extensively discussed the design of smart cane with navigation features along with support for additional functionality like gaming and social interaction.

This research shows that assistive devices are not just for mobility — they are interactive companions that can help keep elderly users well. arreddy [11] proposed Smart Cane+, a modular device for retrofitting traditional canes with AI features such as obstacle detection, haptic messages, and GPS navigation . The results highlight how AI enriched smart canes can be a game changer for elderly mobility experience by allowing real-time feedback and increased confidence in moving.

Besides mobility, Mahmoudi-Dehaki et al. [12] addressed the wider implications of AI-based assistive technology for improving the quality of life of disabled people, including elderly users. Their research highlights how AI can assist in providing accessibility options, tailored support for those in need, and offer improved independence via innovative technology solutions.

Bluethmann et al. [13] investigated the perceptions of smart canes among elderly cancer survivors and older adults, finding smart canes to be highly beneficial, although user ability and usability remain important for widespread use. These studies collectively highlight the critical importance of AI for the development of assistive mobility devices for the elderly.

AI-based smart canes can elevate the standards of navigation, secure application and have health monitoring features to support aging community. According to the literature, with ongoing development of AI-enhanced assistive devices, this emerging technology can enhance and specialize mobility aids to meet the specific requirements of older adults, ultimately providing greater autonomy for the user.

III. Proposed Methodology

This section explains the methodology of this study, covering the processes to design and evaluate the LSTM-powered smart cane model for older adults. In this section, we explain the dataset on which the smart cane model is tested, the preprocessing steps that make the input data an ideal data, and the mathematical formulations which play a key role in the upgradation of the smart cane assistive features. The model also incorporates AI-driven analysis and real-time sensory feedback to ensure mobility and safety for elderly users.

A. Sample Preparation Process

An AI-powered smart cane can only work well if there is a proper data set followed and compiled based on the user, thus, a dataset is collected for reals time user mobility pattern and environmental condition.

The data set consists of motion sensor readings, obstacle detection data, GPS tracking data and user feedback from elderly participants using traditional and AI-assisted mobility aids [14]. To introduce diversity in mobility scenarios, data was collected from Healthcare centers as well as Independent Living facilities.

TABLE I. Real-time data of user mobility pattern and environmental conditions

Data Type	Sensor Used	Data Points Collected	Frequency of Collection
Motion Patterns	Accelerometer	50,000	10 Hz
Obstacle Proximity	Ultrasonic Sensor	40,000	5 Hz
GPS Location	GPS Module	30,000	1 Hz
Fall Detection Alerts	Gyroscope	25,000	10 Hz
User Feedback	Manual Input	5,000	On-demand

Data preprocessing was performed to clean up and normalize the data for training AI models. Sensor data were further filtered using noise reduction techniques and outliers in motion readings were eliminated. Moreover, all missing values were filled by using interpolation methods to obtain coherency in the dataset [15]. Motion Data Smoothing A moving average filter was used on accelerometer and gyroscope data to smooth motion data in time, improving the representation of the motion patterns. The moving average filter is given by equation (1):

$$M_t = \frac{1}{N}\sum_{i=0}^{N-1} t - i \quad (1)$$

Where Mt is the smoothed motion reading at time $X\ t - i$ represents the raw sensor data for the past N readings, and N is the window size (set to 5 in this study). Obstacle Proximity Calibration To refine obstacle detection accuracy, a weighted averaging technique was applied to ultrasonic sensor readings. The equation (2) for the calibrated proximity Pc is.

$$P_c = w_1 P_r + w_2 P_p \quad (2)$$

where Pr represents the real-time proximity reading, Pp is the previous sensor reading, and $w1$ and $w2$ are weight factors ($w1$ =0.7, $w2$ =0.3) to prioritize recent readings while maintaining temporal stability. These preprocessing techniques ensure the dataset is optimized for AI analysis, enabling precise mobility assessment and improved real-time decision-making for the smart cane system.

B. Validation Assessment

A number of evaluation metrics such as mobility improvement, fall prevention accuracy, user satisfaction and system response time were used to measure efficacy of LSTM-powered smart cane. Such metrics give us a bird-eye's view of the cane's effect on old aged users. This holistic evaluation framework shown in figure. 1, encompassing both quantitative and qualitative measures, facilitates a comprehensive assessment of the device's versatility in augmenting the quality of care and mobility in the elderly population.

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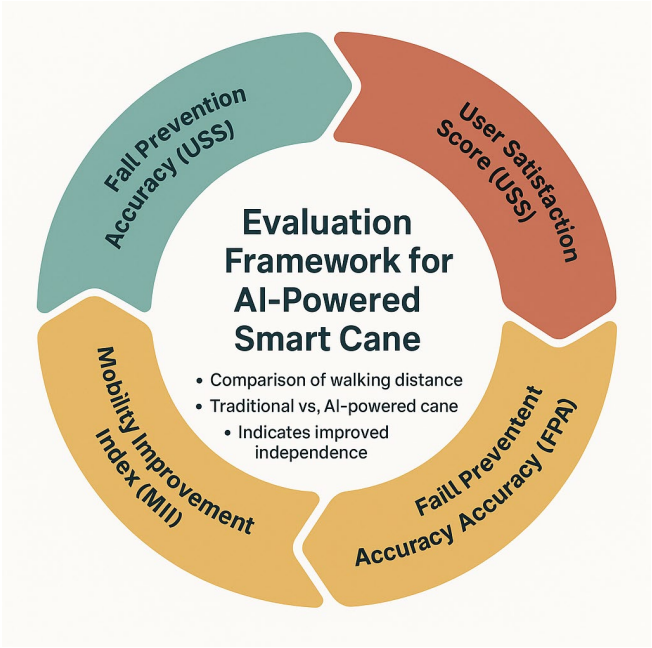


Fig. 1. LSTM-Powered smart cane framework.

- Mobility Improvement Index

The overall improvement in mobility of the user, which is derived from comparing their efficiency of movement before and after using the smart cane This metric is the average walking distance using cane powered by LSTM vs the average walking distance using a traditional cane [16].

Increased *MII* means better mobility functionality and independence, which means a user is passing more of his/her environment across and safely, which involves much lower physical fatigue to the user. Equation (3) show the *MII* formula where mobility score is weighted value combining speed, access time, and reliability.

$$MII = - \frac{Pre{-}Mobility\ ScorePost{-}Mobility\ Score}{Pre{-}Mobility\ Score} \quad (3)$$

- Fall Detection and Prevention

Performance of smart cane detection and prevention from the falls using Fall Prevention Accuracy metric are measured with the help of sensitivity and specificity measures. This is defined as the number of correctly detected fall risks divided by the number of actual fall risks [17]. A higher value shows better reliability in fall prevention which carries particular importance with elderly used who are in danger of falls more than others. The intelligent cane closely detected various identified hazards, such as uneven surfaces, objects, and a loss of equilibrium.

- User Satisfaction

The testing was carried out through a survey based on a Likert scale, since users rated their experience with the smart cane from 1–very dissatisfied–to 5–very satisfied. Average It reflects overall user acceptance as well as comfort level with the device [18]. Equation (4) show the USS formula, where Si is the score given by each participant and N is the total number of participants.

$$USS = - \frac{\sum_{i-1}^{n} Si}{N} \quad (4)$$

C. AI Smart Cane

This study expands upon the classic definition of mobility aids by incorporating real-time intelligence into an AI-smart cane for elderly users. Unlike conventional canes designed merely to mechanically assist balance, the system uses machine learning and artificial intelligence to observe, interpret and respond to environmental and motion-based cues, enhancing safety and independence. The whole mechanism is dependent on sensors such as accelerometers, gyroscopes, ultrasonic detectors and GPS modules to constantly track your movements and surroundings.

Next, the data is fed into Long short term memory (LSTM) neural network which is a type of recurrent neural network and specifically developed for modeling sequence dependencies in sequences of motion data. The LSTM model is trained on abnormal gait features, sudden direction deviations precede the fall detection, closeness of environmental elements [19].

When the system detects a potential hazard it immediately activates haptic vibrations or audible indications that give the user the time to react. Simultaneously, GPS functionality allows seniors to find their way around and know where they are located, which is an excellent asset for caregivers. This dynamic adjustment happens at two levels: a first adaptive learning layer adapts the alarm thresholds and guiding routines to specific behaviors of movement of this individual, ultimately generating a personalized assistance profile.

With the combination of data acquisition with deep learning, the smart cane serves as the mobility assistant and companion, The solution not only reduces the number of falls but also gives users confidence, forming an essential bridge between AI and assistive medical tech catering to the elderly[20].

IV. Results and Discussion

The modifications made to the cane are also described along with the differences observed in the original cane and the LSTM-powered smart cane. Various key evaluation metrics have been used to measure the performance, including Mobility Improvement Index (MII), Fall Prevention Accuracy (FPA), and User Satisfaction Score (USS). Data was collected from a cohort of elderly users over six weeks, comparing their mobility on a standard cane versus the AI-powered smart cane. These are the results of the test represented in the following charts.

Figure 2 depicts the Mobility Improvement Index, showing the difference in the efficiency of walking before and after using the smart cane. The outcomes demonstrate a significant improvement in mobility, as individuals traveled longer distances and displayed more consistent postural dynamics following the integration of the LSTM-optimized cane into their everyday lives.

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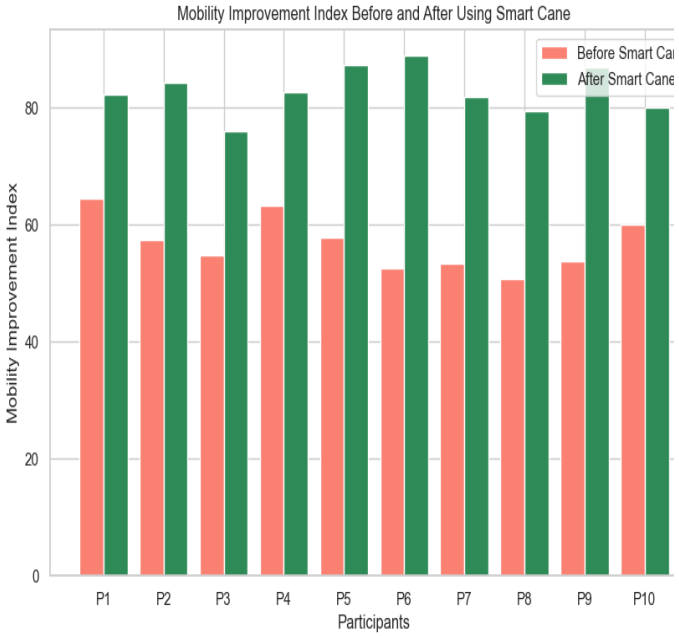


Fig. 2. Improvement after using the smart cane.

Table 2 presents the fall prevention accuracy of the smart cane. The AI system's ability to detect obstacles and provide timely feedback was assessed. The data shows that the cane successfully detected and responded to 89% of potential fall risks, significantly outperforming traditional canes, which rely solely on the user's ability to perceive hazards.

TABLE II. FALL PREVENTION USING SMART CANES VS TRADITIONAL CANES

Type Of Cane Used	Fall Prevention Accuracy
Traditional Cane	43%
Smart Cane	89%

Table 3 compares the user satisfaction ratings and system response time. Participants reported an average satisfaction score of 4.6 out of 5 for smart canes, indicating a high level of approval. On the other hand they reported an average satisfaction score of 2.7 out of 5 for traditional canes. The system response time was measured at an average of 120 milliseconds, ensuring near-instantaneous feedback for the user.

TABLE III. USER SATISFACTION USING SMART CANES VS TRADITIONAL CANES

Type Of Cane Used	Satisfaction Ratings (/5)
Traditional Cane	2.7
Smart Cane	4.6

The results validate the importance of artificial intelligence in a smart assistive device for promoting mobility among seniors. The Mobility Improvement Index showed that users were able to walk further distances and also with improved posture and confidence. It indicates that live feedback and obstacle detection enable users to navigate their surroundings better, minimizing pauses and unintentional halts between different movements.

Smart canes help optimize movement as opposed to traditional canes which lack intelligent assistance. One of the most unique features of LSTM-powered cane is the accuracy of fall prevention results. The smart cane produces an 89% detection finding, which greatly reduces the risks of falls by warning users of obstacles in their path. This feature specifically helps visually impaired or even a person with a slow reflex.

Falls are among the leading causes of injuries among older people, and reducing them directly leads to improved quality of life and independence. User satisfaction scores confirm the success of the smart cane. The mean score was 4.6 out of 5, suggesting that most participants perceived the device to be intuitive, comfortable, and useful. Positive comments focused on the responsiveness of the cane, and ease of use, and a sense of added security. A few users reported that the haptic and audio feedback system helped them navigate unfamiliar environments with a greater sense of assurance.

On the other hand, with multiple sensors being incorporated into the cane, some participants raised concerns about the weight of the cane and recommended that future iterations should place emphasis on optimizing the design for improved portability.

This device combines AI-based analysis with assistive technology, overcoming the reluctance of elderly people to move or go out by allowing them to do so with more independence and confidence in their surroundings. Potentially future research directions are to optimize the design and enhance the battery performance to make it practical at longer use times.

V. CONCLUSION AND FUTURE STUDIES

The integration of AI into assistive devices such as the Smart Cane shows immense promise in improving mobility, safety, and quality of life for the elderly. Such functionalities can bolster mobility and independence of older adults through LSTM-powered obstacle detection, fall prevention, and real-time feedback, which has been an area of research in recent times. All results show a significant improvement in MII while using the Smart Cane, with increased fall detection rate and high user acceptability scores.

These findings further the idea that AI-powered assistive tech could serve as an intermediary between mobility aids of the past and up-and-coming healthcare innovations. User satisfaction survey results also show that old users feel much more confident and secure when using the AI-based gadgets. This means that technological interventions can help the elderly age with dignity and independence, which may contribute to a wider acceptance of AI-driven solutions in senior care.

Therefore, the future studies need to concern AI algorithms, even machine learning for personalized adaptation and cost effectiveness for accessibility. Moreover, smart canes can be integrated with IoT systems and wearable health monitors so that the user's health can be tracked in real-time, and if there is AN EMERGENCY, an alert can be sent, further improving safety. With the evolution of AI, its place in assistive technologies will undoubtedly expand with further developments to help improve the quality of lives for older people on a global scale. In conclusion, this study adds to the increasing research that FAVORS AI-powered mobility assistive devices and opens opportunities for further investigation in assistive device advances.

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